

30 CHAPTERWISE IMPORTANT QUESTIONS

Mathematics Standard (041) | CBSE 2026-27

Introduction to Trigonometry

CBSE Mathematics 2026-27 | Board-examinable

20 numerical / problem-solving questions and 10 theory, proof or method questions. Workings and answers are written for Class X revision.

Question 1

In a right triangle, opposite side = 3 and adjacent side = 4 for angle theta. Find sin theta, cos theta and tan theta.

Answer:

Hypotenuse = 5. $\sin \theta = 3/5$, $\cos \theta = 4/5$, $\tan \theta = 3/4$.

Question 2

In a right triangle, opposite side = 5 and adjacent side = 12 for angle theta. Find sin theta, cos theta and tan theta.

Answer:

Hypotenuse = 13. $\sin \theta = 5/13$, $\cos \theta = 12/13$, $\tan \theta = 5/12$.

Question 3

In a right triangle, opposite side = 8 and adjacent side = 15 for angle theta. Find sin theta, cos theta and tan theta.

Answer:

Hypotenuse = 17. $\sin \theta = 8/17$, $\cos \theta = 15/17$, $\tan \theta = 8/15$.

Question 4

In a right triangle, opposite side = 7 and adjacent side = 24 for angle theta. Find sin theta, cos theta and tan theta.

Answer:

Hypotenuse = 25. $\sin \theta = 7/25$, $\cos \theta = 24/25$, $\tan \theta = 7/24$.

Question 5

In a right triangle, opposite side = 9 and adjacent side = 12 for angle theta. Find sin theta, cos theta and tan theta.

Answer:

Hypotenuse = 15. $\sin \theta = 9/15$, $\cos \theta = 12/15$, $\tan \theta = 9/12$.

Question 6

In a right triangle, opposite side = 12 and adjacent side = 16 for angle theta. Find sin theta, cos theta and tan theta.

Answer:

Hypotenuse = 20. $\sin \theta = 12/20$, $\cos \theta = 16/20$, $\tan \theta = 12/16$.

Question 7

In a right triangle, opposite side = 20 and adjacent side = 21 for angle theta. Find sin theta, cos theta and tan theta.

Answer:

Hypotenuse = 29. $\sin \theta = 20/29$, $\cos \theta = 21/29$, $\tan \theta = 20/21$.

Introduction to Trigonometry - Continued

Question 8

In a right triangle, opposite side = 11 and adjacent side = 60 for angle theta. Find sin theta, cos theta and tan theta.

Answer:

Hypotenuse = 61. $\sin \theta = 11/61$, $\cos \theta = 60/61$, $\tan \theta = 11/60$.

Question 9

In a right triangle, opposite side = 6 and adjacent side = 8 for angle theta. Find sin theta, cos theta and tan theta.

Answer:

Hypotenuse = 10. $\sin \theta = 6/10$, $\cos \theta = 8/10$, $\tan \theta = 6/8$.

Question 10

In a right triangle, opposite side = 10 and adjacent side = 24 for angle theta. Find sin theta, cos theta and tan theta.

Answer:

Hypotenuse = 26. $\sin \theta = 10/26$, $\cos \theta = 24/26$, $\tan \theta = 10/24$.

Question 11

Evaluate $\sin 30 \cos 60 + \cos 30 \sin 60$.

Answer:

Using standard trigonometric values, the value is 1.

Question 12

Evaluate $\sin 45 \cos 45$.

Answer:

Using standard trigonometric values, the value is $1/2$.

Question 13

Evaluate $\tan 45 + \cot 45$.

Answer:

Using standard trigonometric values, the value is 2.

Question 14

Evaluate $\sin 60 / \cos 30$.

Answer:

Using standard trigonometric values, the value is 1.

Question 15

Evaluate $\cos 60 / \sin 30$.

Answer:

Using standard trigonometric values, the value is 1.

Question 16

Evaluate $\tan 30 \times \tan 60$.

Answer:

Using standard trigonometric values, the value is 1.

Introduction to Trigonometry - Continued

Question 17

Evaluate $\sin 30 + \cos 60$.

Answer:

Using standard trigonometric values, the value is 1.

Question 18

Evaluate $\sin 45/\cos 45$.

Answer:

Using standard trigonometric values, the value is 1.

Question 19

Evaluate $\sec 60 - \tan 60$.

Answer:

Using standard trigonometric values, the value is $2 - \sqrt{3}$.

Question 20

Evaluate $\operatorname{cosec} 30 - \cot 45$.

Answer:

Using standard trigonometric values, the value is 1.

Question 21

State the main formula, rule or theorem used in Introduction to Trigonometry.

Answer:

$\sin \theta = \text{perpendicular/hypotenuse}$, $\cos \theta = \text{base/hypotenuse}$, $\tan \theta = \text{perpendicular/base}$.

Question 22

What do the important symbols or quantities in Introduction to Trigonometry represent?

Answer:

The ratios are defined for an acute angle in a right triangle.

Question 23

Write the correct method for solving a Introduction to Trigonometry numerical.

Answer:

Identify sides relative to θ , find a missing side using Pythagoras if needed, then form the ratio.

Question 24

Which unit, sign convention or condition must be checked in a Introduction to Trigonometry answer?

Answer:

Write the given data first, use the stated convention, and include units wherever the final quantity has a unit.

Question 25

Why must the final answer in Introduction to Trigonometry be checked against the given condition?

Answer:

Checking catches wrong signs, arithmetic errors and values that are impossible in the original situation.

Introduction to Trigonometry - Continued

Question 26

What is one common error in Introduction to Trigonometry questions?

Answer:

Students often substitute a value in the wrong formula or skip a condition. Write the formula before substituting.

Question 27

How should full working be shown in a Introduction to Trigonometry numerical?

Answer:

Show the formula or theorem, substitution, simplification, calculation and final answer in that order.

Question 28

When is a diagram, table or graph useful in Introduction to Trigonometry?

Answer:

Use it when it makes the given quantities and their relationship clear before calculation.

Question 29

How can a student decide which formula to use in Introduction to Trigonometry?

Answer:

Identify the quantity asked, list the known quantities, then select the formula or theorem connecting them.

Question 30

What makes a high-scoring answer in Introduction to Trigonometry?

Answer:

Correct reasoning, complete steps, an appropriate formula, a clearly stated final result and a quick verification.

Sources: CBSE Mathematics curriculum 2026-27; CBSE Class X sample papers and marking schemes 2026-27; NCERT textbooks.